PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 10 – Clustering: Distance, Evaluation, and Classical Algorithms

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Three periods: from similarity to useful groups

From Lecture 09

Clustering removes target labels. Define similarity and purpose before interpreting groups.

Period	Questions and calculations
1: 45 min	Objects, scaling, distances, K-means assignments and means, initialization, and K.
2: 45 min	Evaluation, outliers, Gaussian mixtures, a full E/M calculation, and covariance control.
3: 45 min	Linkage, dendrogram cuts, density neighborhoods, graph grouping, and selection.

One shared dataset

Activity scores $X = \{0, 1, 2, 7, 8, 9\}$ support distance, K-means, silhouette, EM, hierarchy, and density calculations.

Next lecture changes the representation through dimensionality reduction. A clean projection must still answer the original grouping question.

Choose one row and meaningful features before choosing an algorithm

Idea

For customer grouping, one row must represent one customer over a stated time window. Aggregate transactions into behavior features; identifiers are keys, not measures of behavioral proximity.

$$\text{transactions} \longrightarrow x_i = (\text{visits}, \text{spend}, \text{recency}) \quad \text{for customer } i$$

Work through it

A customer with ten orders is not ten independent customers. Our six activity scores summarize a deliberately chosen behavior dimension: $0, 1, 2, 7, 8, 9$.

Interpretation

A useful partition should support a question such as outreach frequency or anomaly triage. It should not inherit arbitrary account-number differences.

Aggregate transactions without turning orders into customers

Question and idea

Customers A and B each spend 60 in one window. A has orders 20,30,10; B has one order 60. Compare their purchasing frequency before clustering.

Customer	Order count	Total spend	Mean order
A	3	60	20
B	1	60	60

Calculate and explain

If the grouping question concerns contact frequency, the count distinguishes these customers. If it concerns total contribution, the spend coordinate treats them alike.

Meaning

One row must have a stable unit of analysis and time window. Feature aggregation expresses the question before any clustering optimization begins.

The same points admit multiple useful partitions

Read the figure

The reference places a dendrogram beside its nested groups. It contains the same four points at different resolutions: p_2, p_3 join first, then p_4, and finally p_1.

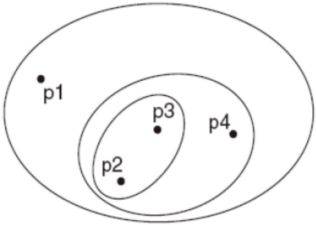
Follow the evidence

A coarse cut gives $\{p_1\}$ and $\{p_2, p_3, p_4\}$; a finer cut separates p_4 from the pair $\{p_2, p_3\}$. Follow the enclosing boundaries and matching tree branches.

Interpretation

Different resolutions answer different questions. A hierarchy preserves these alternatives; it does not establish a universally correct customer taxonomy.





A grouping resolution is a choice about which differences matter

Question and idea

A coarse low/high partition of scores 0,1,2,7,8,9 gives two groups. A finer partition can distinguish the highest-activity rows without changing the input observations.

$$P_2 = \{0, 1, 2 \mid 7, 8, 9\}, \qquad P_3 = \{0, 1, 2 \mid 7, 8 \mid 9\}$$

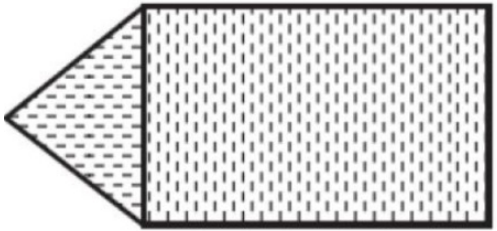
Calculate and explain

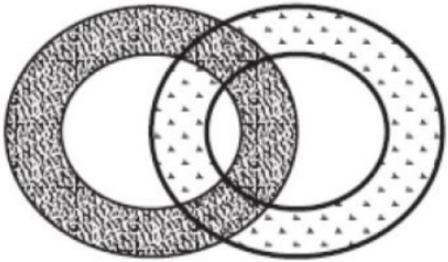
Both describe the same data at different resolution. The extra group is useful only if score 9 represents a distinction supported by repeated evidence and a different intended action.

Meaning

An extra color is not automatically a new customer type. Ask whether the finer group remains stable and has a meaningful original-feature profile.

A grouping resolution is a choice about which differences matter





(e) Conceptual clusters. Points in a cluster share some general property that derives from the entire set of points. (Points in the intersection of the circles belong to both.)

Hard, soft, partial, and nested membership answer different questions

Idea

A hard partition assigns each row to one cluster. Soft probabilistic membership assigns nonnegative weights summing to one. Density clustering may leave noise unassigned; a hierarchy returns nested groups.

$$z_i \in \{1, \ldots, K\} \quad \text{versus} \quad r_i = (0.7, 0.3), \quad \sum_k r_{ik} = 1$$

Work through it

A boundary customer may be 70% compatible with one mixture component and 30% with another under that model. A noise label instead means no accepted dense group.

Interpretation

A responsibility is model-based uncertainty, not the percentage of a person belonging to a social category. Cluster IDs can be permuted without changing the partition.

Soft membership can be counted without inventing fractional people

Question and idea

Three rows have two-component responsibility vectors. Summing one column gives its effective sample count; each row still remains one observed customer.

$$r_1 = (.9, .1), \quad r_2 = (.6, .4), \quad r_3 = (.2, .8)$$

$$N_1 = .9 + .6 + .2 = 1.7, \quad N_2 = .1 + .4 + .8 = 1.3, \quad N_1 + N_2 = 3$$

Calculate and explain

A hard maximum-responsibility assignment would give counts 2 and 1, while the soft description preserves uncertainty about the middle row. Effective counts need not be integers.

Meaning

The responsibilities are posterior weights under a model, not pieces of a person or evidence of simultaneous social-category membership.

Compute how scaling changes the nearest neighbor

Idea

Euclidean distance weights coordinate changes by their numerical units. Standardization uses training or analysis-reference statistics to express changes relative to typical feature variation.

$$A = (1, 100),\ B = (2, 110),\ C = (4, 101)$$

$$d(A, B) = \sqrt{1 + 100} = 10.05; \quad d(A, C) = \sqrt{9 + 1} = 3.16$$

$$s = (1, 100): \quad d_z(A, B) = \sqrt{1 + 0.01} = 1.005; \quad d_z(A, C) = \sqrt{9 + 0.0001} = 3.000$$

Work through it

The closest row changes from C to B when dollars are placed on a deliberate scale. Scaling expresses which kinds of difference should matter.

Interpretation

Changing currency units should not accidentally redefine customer similarity. Robust scales may be preferable with extreme observations.

A robust scale preserves ordinary differences without deleting outliers

Question and idea

For values 1,2,3,4,100, use the median 3 and linear-interpolation quartiles $Q_1 = 2, Q_3 = 4$. A median/IQR transform keeps ordinary differences interpretable while leaving the extreme row visible.

$$z = (x - 3)/(4 - 2) \Rightarrow (-1, -.5, 0, .5, 48.5)$$

Calculate and explain

The first four rows span 1.5 scaled units; the outlier remains far away at 48.5. This transform resists allowing the outlier to set the center and scale, but it does not remove its distance or squared-error influence.

Meaning

Quartile definitions differ for small samples. State the convention, then inspect outlier handling separately from scaling.

Match the metric to the meaning of a feature

Choose the invariance

Smaller distances mean closer rows; larger similarities mean closer rows. The feature meaning determines which comparison is useful.

Quantity	Definition	Worked example
Euclidean distance	$\sqrt{\sum_j (x_j - z_j)^2}$	$(1,2)$ vs $(2,4)$: $\sqrt{5}$
Cosine similarity	$x^T z/(\|x\|\|z\|)$	same vectors: 1
Jaccard similarity	$\|A \cap B\|/\|A \cup B\|$	$\{a,b\}, \{b,c\}$: $1/3$

Interpretation

Proportional text counts share a direction despite different lengths. Shared purchases matter more than the many items neither customer bought.

Categorical IDs have no meaningful numeric gap. Define mixed-feature weights, missing-data handling, and zero-vector or empty-set rules explicitly.

Different norms compare the same coordinate change differently

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Question and idea

For a difference vector $(3, 4)$, compute three common norms. They measure a total coordinate path, straight-line length, and largest coordinate movement respectively.

$$\|\Delta x\|_1 = |3| + |4| = 7, \quad \|\Delta x\|_2 = \sqrt{9 + 16} = 5, \quad \|\Delta x\|_\infty = 4$$

Calculate and explain

A Manhattan-style constraint may fit movements restricted to coordinate paths; Euclidean geometry fits rotations; a maximum norm asks whether any individual coordinate exceeds a tolerance.

Meaning

No norm is universally closer to the truth. Choose the comparison from feature meaning, and keep the objective and prototype update compatible.

Combine numeric and categorical differences explicitly

Question and idea

A simple mixed dissimilarity averages normalized numeric differences and categorical mismatches. This teaching example uses an age range 40 and one product category with equal feature weights.

$$x = (20, \mathrm{A}), \quad z = (40, \mathrm{A}): \quad d = (|20 - 40|/40 + 0)/2 = .25$$
$$z' = (40, \mathrm{B}): \quad d(x, z') = (.5 + 1)/2 = .75$$

Calculate and explain

Changing the product category has a stated cost rather than an arbitrary difference between encoded numbers. Equal weighting is a policy that can be changed with a reason.

Meaning

Missing features alter available comparisons. Record denominators and avoid treating a distance based on one shared field as equally supported as a complete comparison.

Build a distance matrix for the shared six-row dataset

Use the same rows

One-dimensional Euclidean distance is absolute difference. The matrix is symmetric and its diagonal is zero.

$$X = (0, 1, 2, 7, 8, 9), \qquad D = \begin{pmatrix} 0 & 1 & 2 & 7 & 8 & 9 \\ 1 & 0 & 1 & 6 & 7 & 8 \\ 2 & 1 & 0 & 5 & 6 & 7 \\ 7 & 6 & 5 & 0 & 1 & 2 \\ 8 & 7 & 6 & 1 & 0 & 1 \\ 9 & 8 & 7 & 2 & 1 & 0 \end{pmatrix}$$

Read the structure

The closest cross-gap is $d(2, 7) = 5$; every within-triple distance is at most 2. This supports a low/high-activity hypothesis in this coordinate.

Representation matters

Another meaningful feature can change proximity. A gap is evidence for this geometry, not a business label.

Check symmetry, diagonals, and the triangle inequality

Question and idea

A valid metric has a zero self-distance, symmetry, and the triangle inequality. An arbitrary dissimilarity may intentionally relax some properties, but then geometric algorithms need suitable assumptions.

$$d(0,2) = 2 = d(0,1) + d(1,2)$$

$$d(A,B) = 1,\ d(B,C) = 1,\ d(A,C) = 3 \Rightarrow 3 > 1 + 1$$

Calculate and explain

Our activity coordinate satisfies the triangle inequality. The second supplied table violates it: going through B is cheaper than the direct value 3, so it cannot be a metric distance.

Meaning

A symmetric matrix alone does not establish metric geometry, just as symmetry alone did not establish a valid SVM kernel.

K-means searches for a compact partition around K centers

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Idea

Choose K centers and assign every row to exactly one. The squared-error objective measures how far rows are from the center of their assigned group.

$$J = \sum_{i=1}^{N} \sum_{k=1}^{K} r_{ik} \|x_i - \mu_k\|^2, \qquad r_{ik} \in \{0, 1\}, \quad \sum_{k} r_{ik} = 1$$

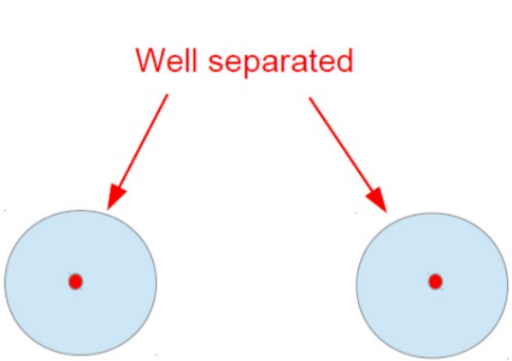
Work through it

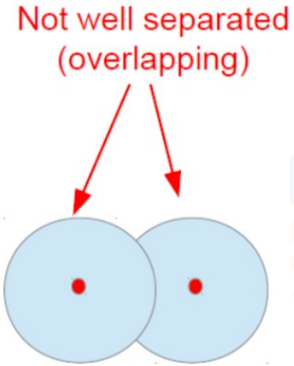
For $X = \{0, 1, 2, 7, 8, 9\}$ and $K = 2$, centers 1 and 8 give squared error $1 + 0 + 1 + 1 + 0 + 1 = 4$.

Interpretation

K-means discovers the best compact groups for this objective, not all possible kinds of structure. The center need not be an observed customer.

Compact clusters are summarized by their centers





Cluster Center

Decompose total variation into within-group and between-group parts

Question and idea

The shared six scores have overall mean 4.5. Under the two triples, within-group SSE is 4 and the group means are 1 and 8. Their separation explains the remaining total variation.

$$T = \sum_i (x_i - 4.5)^2 = 77.5$$

$$W = 4, \qquad B = 3(1 - 4.5)^2 + 3(8 - 4.5)^2 = 73.5, \qquad T = W + B$$

Calculate and explain

The large reduction from one group to two is not mysterious: most variation lies between the two means. The decomposition helps interpret an elbow and dispersion-based internal criteria.

Meaning

This identity depends on arithmetic group means and squared Euclidean error. It is not automatically the objective of density or graph clustering.

Assignment chooses the nearest current center

Idea

With centers fixed, every row minimizes its own contribution independently. Select the center with the smallest squared distance; comparing ordinary distances gives the same winner.

$$z_i = \arg\min_k \|x_i - \mu_k\|^2$$

$$\mu_1 = 0,\ \mu_2 = 7: \quad x = 2 \Rightarrow (4, 25), \quad x = 8 \Rightarrow (64, 1)$$

Work through it

The first assignment is $C_1 = \{0, 1, 2\}$ and $C_2 = \{7, 8, 9\}$. The current dividing location is the midpoint 3.5.

Interpretation

Assignment uses the current centers, not the final centers. Fix a tie rule so repeated equal-distance rows are handled consistently.

Assign new activity scores and define the tie rule

Class exercise

With fitted centers 1 and 8, assign a new score 4 by comparing squared distances. The exact midpoint 4.5 is a tie, even though all original training rows were comfortably separated.

$$x = 4: \quad d_1^2 = 9, \quad d_2^2 = 16 \Rightarrow z = 1$$
$$x = 4.5: \quad d_1^2 = d_2^2 = 12.25$$

Answer and explanation

The new score 4 belongs to the lower-activity center under this rule. At 4.5 use a declared tie policy or flag uncertainty rather than inventing a geometric preference.

Meaning

Assignment of a new row is distinct from refitting the centers using it. Decide whether the model is fixed or an updated analysis is intended.

The mean is the exact update for squared distance

Idea

With assignments fixed, each center solves a separate quadratic problem. Differentiating the sum of squared distances sets the center to the arithmetic mean of its assigned rows.

$$\frac{\partial}{\partial \mu_k} \sum_{i \in C_k} (x_i - \mu_k)^2 = 2 \sum_{i \in C_k} (\mu_k - x_i) = 0$$

$$\mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

Work through it

For our two triples, $\mu_1 = (0 + 1 + 2)/3 = 1$ and $\mu_2 = (7 + 8 + 9)/3 = 8$. Each coordinate is averaged separately for vector data.

Interpretation

The mean follows from the squared-error objective. Changing the distance or loss can require a different prototype update.

Compute a vector center and see why it may be synthetic

Question and idea

Three two-feature customers have coordinates$(1,1),(1,2),(2,1)$. The squared-Euclidean center averages each coordinate; it does not select the observed row nearest the visual middle.

$$\mu = (4/3, 4/3)$$
$$J = (2/9) + (5/9) + (5/9) = 4/3$$

Calculate and explain

The mean is not one of the three rows. If we require an actual customer prototype, that is a medoid requirement and changes the feasible representatives.

Meaning

A synthetic mean can summarize typical behavior, but it should not be described as an observed individual. Compare original units and feature validity when presenting a center.

One complete iteration lowers the objective from 10 to 4

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Idea

Start with centers $0, 7$, assign the two triples, and then update to $1, 8$. Reassign once more: every row remains in its previous triple, so the process has reached a fixed partition.

$$J_{\text{before}} = (0^2 + 1^2 + 2^2) + (0^2 + 1^2 + 2^2) = 10$$
$$J_{\text{after}} = ((-1)^2 + 0^2 + 1^2) + ((-1)^2 + 0^2 + 1^2) = 4$$

Work through it

The decision midpoint moves from 3.5 to 4.5, but no row crosses it. The updated centers describe low and high activity in the original coordinate.

Interpretation

Stopping requires stable assignments, sufficiently small center movement, or a small objective improvement—not merely one attractive plot.

A stable partition can still be a worse local optimum

Class exercise

Use four points$(0,0),(0,2),(4,0),(4,2)$ and $K = 2$. Both horizontal and vertical partitions have self-consistent mean centers, but their objectives differ sharply.

$$\text{horizontal means } (2,0),(2,2): \quad J = 4+4+4+4 = 16$$
$$\text{vertical means } (0,1),(4,1): \quad J = 1+1+1+1 = 4$$

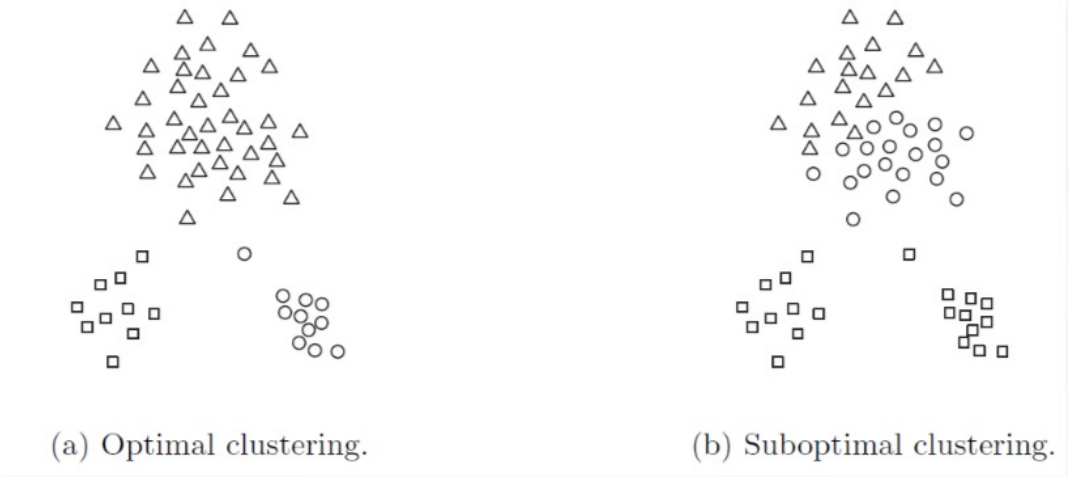
Answer and explanation

Reassignment preserves each partition because each point is nearer its own corresponding center. Convergence alone cannot choose between the stable SSE 16 and SSE 4 solutions.

Meaning

Repeated starts target optimization reliability. They do not prove a unique meaningful grouping or eliminate all structural bias.

A stable partition can still be a worse local optimum



(a) Optimal clustering.

(b) Suboptimal clustering.

Read the original iteration figure as alternating decisions

Read the figure

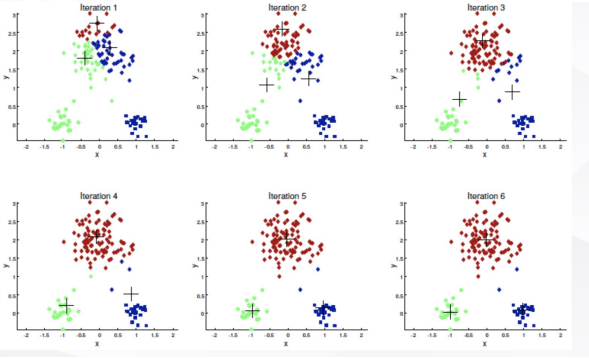
Read the six panels in temporal order. Colored points indicate assignments and crosses indicate current centers. First identify moved centers, then inspect rows whose nearest center changes.

Follow the evidence

A displaced center can pull several rows across a boundary; their changed membership then shifts the next mean. The last panels stabilize because both decisions stop changing appreciably.

Interpretation

Each exact assignment or mean update cannot increase SSE. Different starting points can still reach different local minima.



Read repeated-start objectives without mistaking label swaps for instability

Question and idea

Four runs on the rectangle end with SSE 4,16,4,4. Compare the objective and the actual partition; swapping cluster IDs does not change any co-membership.

$$\text{run objectives} = (4, 16, 4, 4)$$

$$\{\text{left} : 1, \text{right} : 2\} \equiv \{\text{left} : 2, \text{right} : 1\}$$

Calculate and explain

The SSE 16 run is a worse local fit. Two SSE 4 runs with swapped IDs describe the same vertical grouping; counting raw ID changes would incorrectly report every row as unstable.

Meaning

Keep the chosen initialization policy, number of starts, objective distribution, and membership comparison. A seed alone does not describe the entire experiment.

Spread seeds with K-means++ and still repeat the run

Idea

After choosing the first seed, sample the next row with probability proportional to its squared distance from the nearest chosen seed. This favors uncovered regions without always choosing the farthest outlier.

$$P(x_i \text{ next}) = \frac{D(x_i)^2}{\sum_j D(x_j)^2}$$

$$\text{seed } 0: \quad D^2 = (0, 1, 4, 49, 64, 81), \quad \sum D^2 = 199$$

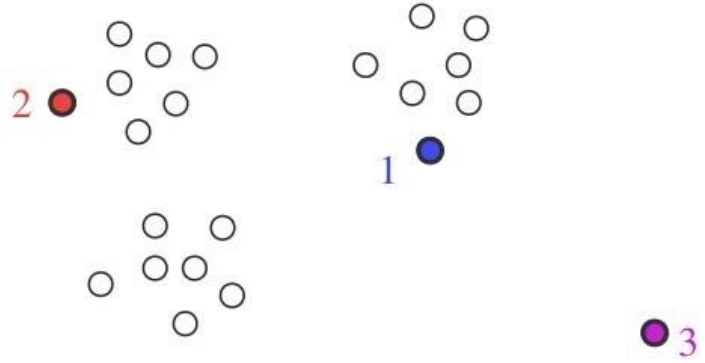
Work through it

Score 9 has probability $81/199 \approx 0.407$ of being selected next. K-means++ is a randomized initialization, not a proof of the globally best partition.

Interpretation

Use explicit repeated starts, compare final SSE and membership stability, and record the seed for reproducibility.

Spread seeds with K-means++ and still repeat the run



Update K-means++ probabilities after a second seed

Question and idea

On the six scores, suppose seeds 0 and 7 are already selected. For a third seed, squared distance is measured to the nearest chosen seed, not only to the most recent one.

$$D^2 = (0, 1, 4, 0, 1, 4), \quad \sum_i D_i^2 = 10$$

$$P(2 \text{ next}) = .4, \quad P(9 \text{ next}) = .4, \quad P(1) = P(8) = .1$$

Calculate and explain

The probabilities now focus on the uncovered ends of each triple. Selected seeds have probability zero. Choosing three centers does not establish that three customer types exist.

Meaning

Initialization uses coverage conditional on the requested K. The decision about K still requires an independent reason and evaluation.

Handle empty groups and stopping conditions explicitly

Idea

An empty group has no mean. Reinitialize its center using a high-error row or split a populated group, then resume the same assignment/update cycle. State the rule rather than silently dividing by zero.

$$\max_k \|\mu_k^{(t+1)} - \mu_k^{(t)}\| < \delta \quad \text{or} \quad \frac{|J_{t+1} - J_t|}{\max(1, J_t)} < \delta$$

Work through it

Our centers stop at $1, 8$ because the next assignments and means are unchanged. At very large N, mini-batch updates approximate this cycle with small samples of rows.

Interpretation

A numerical stop or iteration cap is different from proving a globally optimal partition. Explain both the final objective and the resulting groups.

Choose K using diminishing returns and usefulness

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Idea

For the best achievable K-means objective, increasing K cannot make SSE larger: extra centers can reproduce a coarser solution. Training SSE therefore rewards ever finer partitions.

$$X = \{0, 1, 2, 7, 8, 9\}: \quad J_1 = 77.5, \quad J_2 = 4, \quad J_3 = 2.5, \quad J_6 = 0$$

Work through it

The improvement from one to two groups is 73.5, while two to three gains only 1.5. This suggests an elbow at two for this coordinate, but business resolution may still differ.

Interpretation

Combine an elbow with silhouette, repeated-start stability, and interpretable profiles. There may be no clear elbow in real data.

An explicit group cost turns diminishing returns into a decision

Class exercise

As a teaching decision rule, penalize each group by a stated cost τ. On the six scores, compare $J_K + \tau K$ instead of choosing the smallest training SSE.

$$J_1 = 77.5,\ J_2 = 4,\ J_3 = 2.5$$
$$\tau = 2: \quad (79.5, 8, 8.5) \Rightarrow K = 2$$

Answer and explanation

Between two and three groups, the SSE gain is 1.5 while the added group costs 2. The cost rule prefers two. A different justified cost can make the finer resolution worthwhile.

Meaning

This is an explicitly constructed utility criterion, not a universal clustering theorem. Formal information criteria and task constraints need their own assumptions.

A compact-center objective has geometric biases

Read the figure

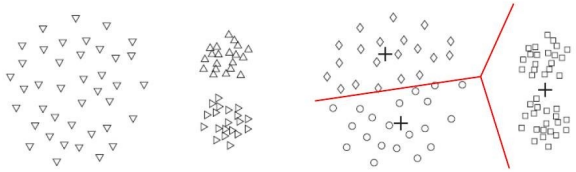
Nearest-center assignments create convex Voronoi regions. K-means therefore struggles with curved groups, strongly unequal sizes, and strongly unequal spreads or densities.

Follow the evidence

Read where a large or sparse true-looking group is split, and where parts from different shapes are joined. Those errors follow from the objective, not necessarily from insufficient iterations.

Interpretation

More starts can improve optimization but cannot change the model geometry. Consider another representation or method when the desired group is nonconvex.



(a) Original points.

(b) Three K-means clusters.

An outlier separates mean, median, and medoid

Idea

A mean minimizes squared error; a coordinate median minimizes one-dimensional absolute error; a medoid is an observed row minimizing total chosen dissimilarity to its group.

$$C = \{0, 1, 2, 30\}: \quad \text{mean} = 8.25, \quad \text{median} = 1.5$$

$$\sum_{x \in C} |x - 1| = 31, \qquad \sum_{x \in C} |x - 2| = 31$$

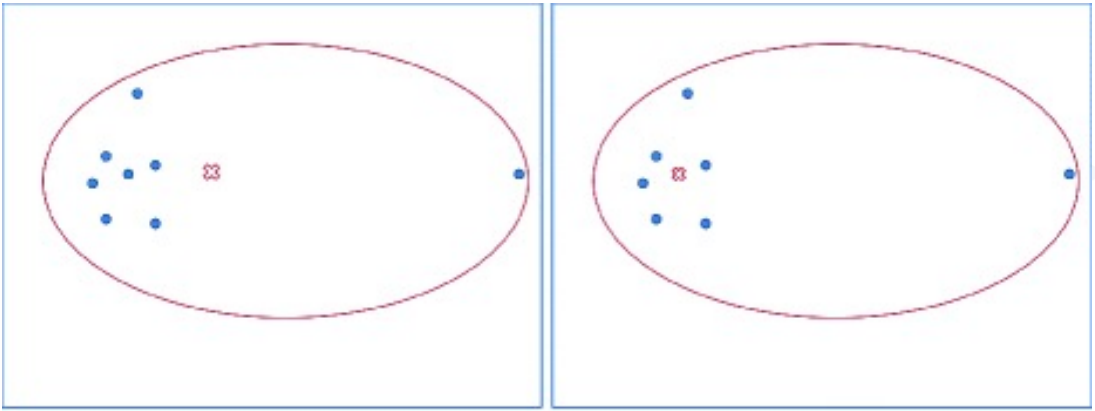
Work through it

The outlier pulls the mean far from the first three rows. Both 1 and 2 are valid medoids for absolute distance; the median 1.5 is not an observed row.

Interpretation

K-medoids can use broader distances and be more robust, but it is not simply the median and often costs more computation.

An outlier separates mean, median, and medoid



(a) Mean (b) Medoid

A medoid depends on the chosen dissimilarity

Question and idea

For $C = \{0, 1, 2, 30\}$, compare an observed representative under absolute and squared distances. The feasible representative set is the same, but the objective is not.

$$\sum |x - 1| = 31 = \sum |x - 2|$$

$$\sum (x - 0)^2 = 905, \quad \sum (x - 1)^2 = 843, \quad \sum (x - 2)^2 = 789$$

Calculate and explain

Absolute-distance medoids 1 and 2 tie; squared-distance medoid 2 is preferred. The unconstrained arithmetic mean 8.25 has SSE 632.75, but it is not an observed row.

Meaning

Requiring an observed prototype does not automatically remove outlier sensitivity from every dissimilarity. State both the representative restriction and the penalty.

Ask three different questions about clustering quality

Three evidence questions

Agreement with reference labels, separation in the chosen geometry, and stability under perturbation are different claims.

Evidence	Question	Example
External	Does grouping match a stated reference?	Adjusted Rand
Internal	Are groups separated in this distance?	Silhouette
Stability	Do reasonable samples and seeds preserve it?	Pair agreement

Customer interpretation

Past response labels may supply a reference, but activity grouping may serve a different purpose. Read each group in original visits, spend, and recency units.

One high score cannot establish useful segments. Name the question and retain evidence relevant to the intended action.

Calculate one silhouette value from the shared distances

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Idea

For a row, a is its mean distance to other members of its own group. b is the smallest mean distance to another group. Compare both using the same metric.

$$s = \frac{b - a}{\max(a, b)}$$

$$x = 1: \quad a = (1 + 1)/2 = 1, \quad b = (6 + 7 + 8)/3 = 7 \quad \Rightarrow \quad s = 6/7 \approx 0.857$$

Work through it

A value near 1 means separated relative to its own spread; near 0 suggests a boundary row; negative means another group is closer on average.

Interpretation

Silhouette favors structures well separated in the chosen distance. It can penalize meaningful curved or variable-density groups.

A misplaced row can have a negative silhouette

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Class exercise

Intentionally place activity score 2 with 7,8,9, leaving 0,1 in the other group. Use the same distance matrix to compute whether the row agrees with its assigned group.

$$a(2) = (5 + 6 + 7)/3 = 6, \qquad b(2) = (2 + 1)/2 = 1.5$$
$$s(2) = (1.5 - 6)/6 = -.75$$

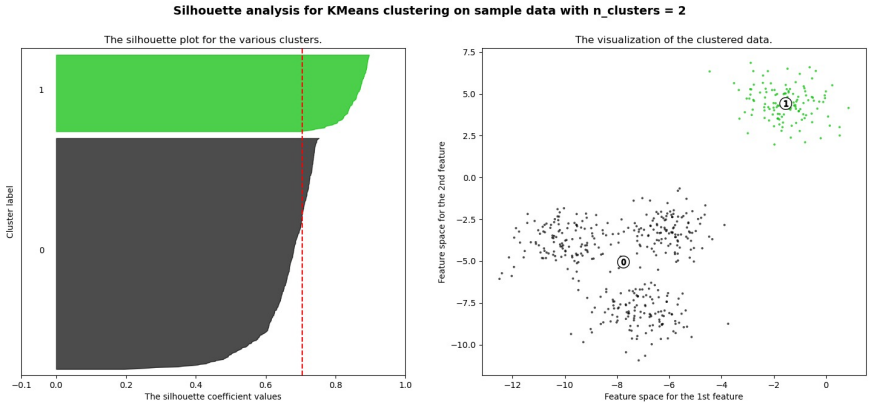
Answer and explanation

The other group is much closer on average. The negative value flags this assignment under the chosen metric, unlike the earlier positive 0.75 when 2 belonged with 0,1.

Meaning

A negative silhouette is a reason to inspect a row and representation. It is not permission to delete inconvenient observations just to raise the score.

Silhouette compares within-cluster and neighboring-cluster distances



Read a silhouette profile, not only its average

Read the figure

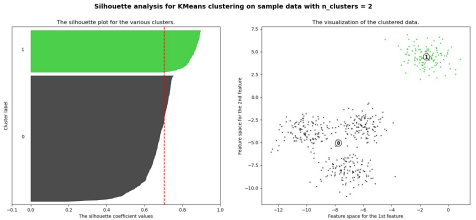
A silhouette plot displays individual rows sorted within each group. Group thickness represents size; negative bars identify rows whose average distance favors another group.

Follow the evidence

Compare whether every group has positive widths, whether one group is mostly near zero, and whether a high overall average is driven by one large easy group.

Interpretation

Use the plot to locate observations for interpretation. Do not rename or delete difficult rows merely to make the score prettier.



Compute the full silhouette average and retain row-level evidence

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Question and idea

Under the original two triples, the six point silhouettes are symmetric. Average them only after calculating the own-group and nearest-other-group distances for each row.

$$s = (.8125, .85714, .75, .75, .85714, .8125)$$

$$\bar{s} \approx .80655$$

Calculate and explain

The inner gap-adjacent scores 2 and 7 have smaller silhouettes than the central scores 1 and 8. The positive mean summarizes clean separation in this coordinate but hides these position differences.

Meaning

A large easy group can dominate an overall average in unbalanced data. Inspect the profile and group sizes as well as the mean.

Compare pairs rather than arbitrary cluster numbers

Idea

Two partitions agree on a pair if both place it together or both separate it. This is invariant to renaming cluster IDs. Adjusted Rand index subtracts expected chance agreement.

$$\mathrm{RI} = \frac{\text{same in both} + \text{different in both}}{\binom{N}{2}}$$

$$A = \{12|34\}, \quad B = \{123|4\}: \quad \text{agreements} = 1 + 2 = 3, \quad \mathrm{RI} = 3/6 = 0.5$$

Work through it

Pair $(1, 2)$ stays together; pairs $(1, 4)$ and $(2, 4)$ stay separated. The remaining three pairs disagree.

Interpretation

Purity can reach 1 by giving every row a singleton group. Compare it with K and chance-adjusted or pair-based evidence.

Calculate an adjusted Rand example from a contingency table

Class exercise

Keep predicted triples of size 3,3, but reference labels have sizes 2,4. The contingency table is $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$.

Compute chance adjustment from pair counts.

$$\sum_{ij} \binom{n_{ij}}{2} = 4, \quad \sum_{i} \binom{a_i}{2} = 6, \quad \sum_{j} \binom{b_j}{2} = 7, \quad \binom{6}{2} = 15$$

$$\mathrm{ARI} = \frac{4 - 6 \cdot 7/15}{(6+7)/2 - 6 \cdot 7/15} = \frac{1.2}{3.7} \approx .324$$

Answer and explanation

The ordinary Rand agreement is 10/15, about 0.667, but its chance-adjusted counterpart is 0.324. Label permutation has no effect on either calculation.

Meaning

Chance correction answers a specific reference-agreement question. Neither number establishes that the reference labels express the intended grouping purpose.

Describe groups in original units and test stability

Idea

Repeat reasonable perturbations: seeds, row resamples, adjacent time windows, and defensible scales. Compare co-membership or align labels before reporting changes.

Group	Description	Evidence to retain
Low activity	scores $0, 1, 2$; center 1	size, original visits, recency
High activity	scores $7, 8, 9$; center 8	size, spread, response patterns

Work through it

A stable low/high division can still be useless if it does not support different actions. A fragile boundary customer need not invalidate the broad pattern.

Interpretation

Keep uncertainty visible; separate a descriptive behavior group from a causal or value judgment about customers.

Measure a changed partition using co-membership

Question and idea

Compare $P = \{0, 1, 2 \mid 7, 8, 9\}$ with $Q = \{0, 1 \mid 2, 7, 8, 9\}$. One row changed sides, but several pair relations changed with it.

$$\text{same pairs in } P = 6, \quad \text{in } Q = 7, \quad \text{shared same} = 4$$

$$\text{shared different} = 15 - 6 - 7 + 4 = 6, \quad \mathrm{RI} = 10/15 \approx .667$$

Calculate and explain

The row at 2 no longer pairs with 0,1 and now pairs with 7,8,9. A simple one-of-six switch statistic and a pair-agreement statistic measure different aspects of stability.

Meaning

Use a declared comparison and perturbation policy. Raw cluster ID agreement would still be invalid when numbering changes.

A Gaussian mixture models density before assigning groups

Idea

Assume a latent component z generates each row: first select component k with weight π_k, then draw the row from its Gaussian distribution.

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k), \quad \pi_k \geq 0, \quad \sum_k \pi_k = 1$$

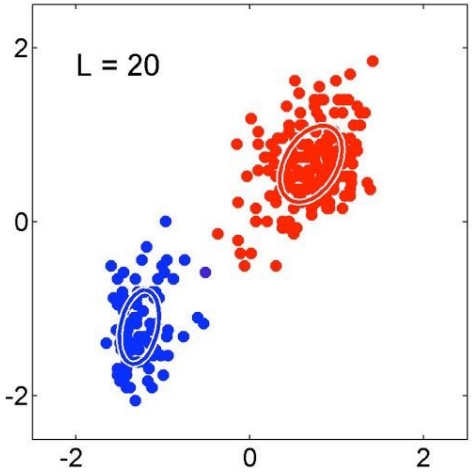
Work through it

Unlike K-means, components may have different spreads, orientations, and population weights. A row in an overlap can receive uncertain posterior membership.

Interpretation

A component is a model of a distribution, not necessarily a one-to-one real customer type. Multiple components can approximate one complicated density.

A Gaussian mixture models density before assigning groups



Mean, covariance, and weight play distinct roles

Idea

The mean gives a component location. Covariance gives within-component spread and orientation. The mixing weight gives how often that component is selected before seeing a row.

$$\mu = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 4 & 1.5 \\ 1.5 & 1 \end{pmatrix}, \quad \pi = 0.3$$

Work through it

A full covariance can describe tilted ellipses; diagonal covariance keeps axes aligned; spherical covariance shares one spread within a component. None of these weights is a customer-specific responsibility.

Interpretation

A broad component can explain a farther row better than a narrow nearby component. Nearest mean alone is not the GMM assignment rule.

Count the parameters before choosing covariance complexity

Question and idea

For $K = 3$ components in $D = 2$ dimensions, means contribute 6 parameters and weights contribute 2. Covariance choices add different numbers of free values.

Covariance	Free covariance values	Total p
Full	$3 \times 3 = 9$	17
Diagonal	$3 \times 2 = 6$	14
Spherical per component	$3 \times 1 = 3$	11

Calculate and explain

Full two-dimensional covariance needs two variances and one cross-covariance per component. The weight sum-to-one constraint removes one weight degree of freedom.

Meaning

The richest shape is not free. Small effective component sizes make its additional parameters difficult to estimate reliably.

The E step converts component densities into responsibilities

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Idea

Given current parameters, Bayes rule assigns normalized posterior weights to each row. These responsibilities sum to one across components for that row.

$$r_{ik} = P(z_i = k \mid x_i) = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}$$

Work through it

If weighted densities are 0.12 and 0.03, responsibilities are 0.8 and 0.2. Their ratio reflects both component shape and prior mixture weight.

Interpretation

E means expectation of hidden membership indicators. It estimates missing assignments under the current model; it does not observe true cluster labels.

A prior mixture weight can outweigh nearest-mean distance

Question and idea

At input 2, compare Gaussians with means 0 and 3, equal variance 1, and prior weights 0.9,0.1. The common normalizing constant cancels when calculating responsibilities.

$$u_1 = .9e^{-2} \approx .1218, \quad u_2 = .1e^{-.5} \approx .06065$$
$$r_1 = u_1/(u_1 + u_2) \approx .668, \quad r_2 \approx .332$$

Calculate and explain

The point is closer to mean 3, yet the much larger prior weight makes component 1 more probable under this supplied model. Distance alone would choose differently.

Meaning

This is a model-dependent posterior, not proof that the more common real-world group must always own ambiguous observations.

Compute an E step for the same six activity scores

Idea

Use two equal-weight one-dimensional Gaussians with initial means $1, 8$ and variances $9, 9$. The first responsibility has a simple logistic expression after canceling common constants.

$$r_{i1} = \frac{1}{1 + \exp\left(7(x_i - 4.5)/9\right)}, \qquad r_{i2} = 1 - r_{i1}$$

x_i	0	1	2	7	8	9
r_{i1}	.971	.938	.875	.125	.062	.029

Work through it

At score 2, the model assigns about 0.875 responsibility to the lower-activity component, while retaining 0.125 for the broad upper component.

Interpretation

These numbers follow from explicitly stated parameters. They are not arbitrary soft labels or observed probabilities from customers.

An outlier can have near-certain responsibility and tiny density

Question and idea

Reuse initial means 1,8, variances 9, equal weights, but query activity 20. The nearer component dominates the posterior even though both component densities are small.

$$r_1(20) = 1/(1 + e^{7(20-4.5)/9}) \approx 5.81 \times 10^{-6}$$
$$r_2 \approx .999994, \qquad p(20) \approx 2.23 \times 10^{-5}$$

Calculate and explain

The model is nearly certain which of its two components is relatively more plausible, while still assigning very low density to the observation as a whole.

Meaning

Responsibility confidence is conditional on the modeled components. For anomaly interpretation, inspect likelihood or density and relevant data-range evidence as well.

The M step updates weighted counts, means, and variances

Idea

Hold the E-step responsibilities fixed. Treat each row as a fractional contribution to each component, then maximize the expected complete-data log likelihood.

$$N_k = \sum_i r_{ik}, \quad \pi_k^{\text{new}} = N_k / N, \quad \mu_k^{\text{new}} = \frac{\sum_i r_{ik} x_i}{N_k}$$

$$\Sigma_k^{\text{new}} = \frac{\sum_i r_{ik} (x_i - \mu_k^{\text{new}})(x_i - \mu_k^{\text{new}})^T}{N_k}$$

Work through it

Responsibilities 0.8 and 0.2 make one row count as 0.8 of a row for the first mean and 0.2 for the second. This is a weighted average, not random fractional data duplication.

Interpretation

Use the newly updated mean inside each covariance update. The row probabilities remain fixed during this M step.

Update mixture proportions after adding one observation

Class exercise

The original six-row E step has effective counts 3 and 3. Add a row at 0 under the same current parameters; its first responsibility is 0.970688.

$$N_1 = 3 + .970688 = 3.970688, \quad N_2 = 3 + .029312 = 3.029312$$
$$\pi_1^{\text{new}} = 3.970688/7 \approx .56724, \qquad \pi_2^{\text{new}} \approx .43276$$

Answer and explanation

The added row does not simply make a hard four-to-three split: it contributes small weight to the other component too. The revised weights still sum to 1.

Meaning

These proportions estimate modeled source frequency under the chosen sample. Biased customer sampling would bias them; EM does not correct representation automatically.

Finish the hand iteration and interpret the parameter movement

Idea

Use the unrounded E-step values from the previous table. Symmetry gives effective counts $N_1 = N_2 = 3$, so both mixture weights remain $1/2$.

$$\mu_1^{\text{new}} = \frac{\sum_i r_{i1} x_i}{3} \approx 1.440, \qquad \mu_2^{\text{new}} \approx 7.560$$

$$\sigma_1^{2\,\text{new}} = \frac{\sum_i r_{i1} (x_i - 1.440)^2}{3} \approx 3.556, \quad \sigma_2^{2\,\text{new}} \approx 3.556$$

Work through it

The broad initial variances shrink from 9 to about 3.556. The means move inward from $1, 8$ because each component still receives small contributions from the opposite triple.

Interpretation

Run the E step again using these new parameters. A complete EM iteration is a parameter-dependent reassignment followed by a weighted update.

A second E step sharpens the shared six-point example

Question and idea

After the first M step, means are 1.440445 and 7.559555, equal variances 3.555790 and equal weights. Recompute responsibility for the same score 2 rather than retaining its old 0.874840.

$$r_1(2) = \frac{1}{1 + \exp((7.559555 - 1.440445)(2 - 4.5)/3.555790)} \approx .986642$$

Calculate and explain

The variance contraction makes score 2 much more strongly associated with the lower component. The updated means also enter the exponent; every E step uses the current full parameters.

Meaning

This one-row change is not a global-convergence proof. Repeat all responsibilities and parameter updates, monitoring the same observed-likelihood objective.

EM raises observed likelihood through a tight lower bound

Idea

Introduce a distribution $q(z)$ over hidden assignments. Jensen uses the concavity of the logarithm to lower-bound observed log likelihood; the E step makes this bound tight at the current parameters.

$$\log p(X \mid \theta) \geq \mathbb{E}_q[\log p(X, z \mid \theta)] + H(q) = \mathcal{L}(q, \theta)$$
$$q^{\text{new}} = p(z \mid X, \theta^{\text{old}}), \quad \theta^{\text{new}} = \arg\max_{\theta} \mathcal{L}(q^{\text{new}}, \theta)$$

Work through it

The M step raises the fixed bound, so observed likelihood cannot decrease for exact EM updates. This does not prove the best possible mixture was found.

Interpretation

Log is concave, not convex. Initialization, singular covariance, and local maxima still matter.

A numerical Jensen bound explains the E step

Question and idea

For one row, weighted component densities are $u = (.12, .03)$. A trial distribution $q = (q_1, q_2)$ has positive entries summing to one. Concavity of log gives a lower bound on the observed log density.

$$\log(u_1 + u_2) \geq \sum_{k=1}^{2} q_k \log \frac{u_k}{q_k}$$

$$q = (.5, .5): \quad -1.89712 \geq \tfrac{1}{2}\log(.24) + \tfrac{1}{2}\log(.06) = -2.12026$$

Calculate and explain

Posterior $q = (.8, .2)$ makes both ratios $u_k/q_k = .15$, so the bound equals $\log(.15)$. The E step selects this tight bound before the M step raises it.

Meaning

Log is concave. Tightness at the current parameters explains monotonic improvement, while global optimality remains unproved.

Verify the observed likelihood increase numerically

Question and idea

For the six activity scores, evaluate the mixture density sum before taking logs, then sum over rows. Use the same model family and dataset for both iterations.

$$\ell(\theta) = \sum_{x \in X} \log \left[\tfrac{1}{2}\mathcal{N}(x \mid \mu_1, v) + \tfrac{1}{2}\mathcal{N}(x \mid \mu_2, v) \right]$$

$$\ell(1, 8, 9) \approx -16.03216, \qquad \ell(1.440445, 7.559555, 3.555790) \approx -14.17178$$

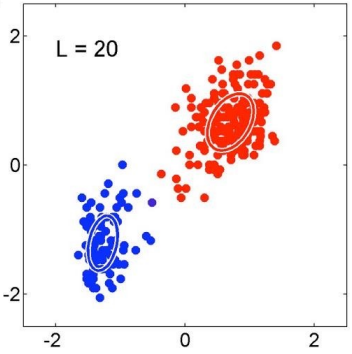
Calculate and explain

The log likelihood rises by about 1.86038: the less-negative number is larger. It is the observed likelihood, not a single component likelihood or the sum of responsibility maxima.

Meaning

This checks one exact teaching iteration. It supports monotonic improvement here, while initialization and possible local optima remain relevant.

Read fitted mixture ellipses as probability models



Read the figure

The original reference shows a fitted mixture after 20 iterations. Read its component means, ellipse orientation, spread, and overlap as parts of a probability model.

Follow the evidence

The two fitted ellipses follow differently located clouds. A hard color keeps only the largest responsibility; a row can remain partly compatible with another component even if its color is unchanged.

Interpretation

The visible ellipses summarize this fitted run. They do not establish a unique population decomposition or show every intermediate E/M update.

Prevent tiny covariance from producing a false victory

Idea

A component can concentrate around one observation with variance approaching zero, driving Gaussian density at that row upward. Unregularized mixture likelihood can therefore be unbounded.

$$\mathcal{N}(\mu \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \longrightarrow \infty \quad \text{as } \sigma^2 \to 0$$

Work through it

Use a covariance floor such as $\Sigma_k + \eta I$, sensible component counts, minimum effective sizes, and repeated initializations. Compare held-out density or task evidence as appropriate.

Interpretation

A very high training likelihood may mean collapse rather than a meaningful new group. Check component size and covariance eigenvalues.

Covariance regularization has a measurable coordinate scale

Question and idea

In one dimension, an almost-collapsed variance 10^{-8} has standard deviation 10^{-4}. Adding a variance regularizer of $\eta = .01$ prevents that component from concentrating on an almost exact point.

$$v^{\mathrm{reg}} = 10^{-8} + .01 \approx .01000001$$

$p(\mu)$ changes from 3989.4 to about 3.9894

Calculate and explain

The regularized standard deviation is about 0.1 in the current coordinate. Whether that regularizer is sensible depends on feature units and the measurement spread.

Meaning

Regularization controls a numerical and statistical pathology; it does not establish that every fitted component has adequate effective size or task meaning.

Choose mixture complexity and connect it back to K-means

Idea

Mixture complexity depends on both K and covariance structure. Information criteria balance fitted likelihood against parameter count; smaller AIC or BIC is preferred under their assumptions.

$$\mathrm{AIC} = 2p - 2\log \widehat{L}, \qquad \mathrm{BIC} = p\log N - 2\log \widehat{L}$$

$$\pi_k = 1/K, \quad \Sigma_k = \sigma^2 I, \quad \sigma^2 \to 0 \Rightarrow \text{nearest-mean responsibilities become hard}$$

Work through it

The limiting assignments connect GMM to K-means. Finite-width mixtures retain uncertainty and can estimate unequal covariance shapes.

Interpretation

A criterion-selected density model need not be the most useful segmentation. Compare component interpretation and stability as well.

Calculate BIC tradeoffs instead of selecting the largest likelihood

Question and idea

For 100 rows, compare candidate models with 5 or 8 free parameters and the stated fitted log likelihoods. All numbers are constructed for this model-selection calculation.

Model	p	$\log \widehat{L}$	$p \log 100 - 2 \log \widehat{L}$
A	5	-100	223.03
B	8	-95	226.84
C	8	-90	216.84

Calculate and explain

B gains 5 likelihood units but pays three extra parameter penalties, so A has lower BIC. C gains 10 units and overcomes that penalty. Lower BIC is preferred under the criterion.

Meaning

BIC evaluates a stated probabilistic model and parameterization. A selected density decomposition still requires stability and useful-group interpretation.

A hierarchy keeps several resolutions in one tree

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Idea

Agglomerative clustering begins with singleton groups and repeatedly merges the closest two. Divisive clustering starts with one group and splits it. The tree records a sequence, not one fixed K.

$$\{0|1|2|7|8|9\} \longrightarrow \{01|2|78|9\} \longrightarrow \{012|789\} \longrightarrow \{012789\}$$

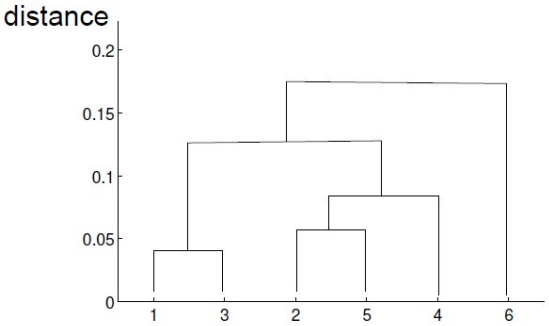
Work through it

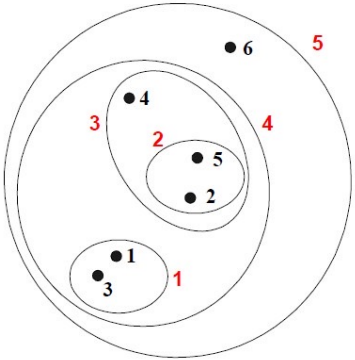
At a coarse resolution our six rows form low/high activity groups; at a finer resolution there may be pairs and singletons. A cut chooses the resolution after the tree is built.

Interpretation

Specify linkage before merging. Different definitions of distance between groups generate different trees from the same pairwise distances.

A hierarchy keeps several resolutions in one tree





Cut heights select different resolutions of the same hierarchy

Class exercise

For the six scores, single linkage merges each triple at height 1 and the two triples at 5. Choose cuts that avoid equalities.

Cut height	Groups	Partition
.5	6	Each row alone
1.5	2	$\{0, 1, 2 \mid 7, 8, 9\}$
5.5	1	All six rows together

Answer and explanation

Cuts 1.5 and 3 give the same groups: no merge occurs between them. The hierarchy records merge events rather than a new partition at every height.

Meaning

Resolution is selected after building the tree, but its meaning still depends on the chosen metric, linkage, and task.

Linkage defines the distance between two multi-row groups

Idea

Single linkage uses the nearest cross-pair, complete linkage the farthest, and average linkage the mean of all cross-pair distances. These definitions emphasize different structures.

$$A = \{0, 1, 2\},\ B = \{7, 8, 9\}$$

$$d_{\text{single}} = 5, \quad d_{\text{complete}} = 9, \quad d_{\text{average}} = \frac{7 + 8 + 9 + 6 + 7 + 8 + 5 + 6 + 7}{9} = 7$$

Work through it

Single linkage can follow chains; complete linkage favors bounded diameter; average linkage compromises. The same groups have three different merge heights.

Interpretation

A merge height is a linkage-specific quantity. Do not compare trees as if their vertical units represented identical criteria.

Average linkage must weight merged groups by their sizes

Question and idea

Let $A = \{0, 1\}$, $B = \{2\}$ and $D = \{7, 8, 9\}$. The all-cross-pair average from a merged group is a size-weighted combination of the old averages.

$$d(A, D) = 7.5, \quad d(B, D) = 6$$

$$d(A \cup B, D) = \frac{2(7.5) + 1(6)}{3} = 7 \neq \frac{7.5 + 6}{2} = 6.75$$

Calculate and explain

A contributes twice as many cross pairs as B because it has two rows instead of one. Equal averaging of the two group-level distances defines a different linkage.

Meaning

Name the averaging convention. The word average alone is insufficient when updating clusters of different sizes.

Work through agglomeration using the shared distance matrix

Idea

Use single linkage and a stated tie rule. The smallest original distance is 1, occurring between several adjacent scores. Merge one tied pair, then update group distances using the minimum cross-pair.

$$\{0\}, \{1\} \to \{0, 1\} \quad \text{at height } 1$$
$$d(\{0, 1\}, \{2\}) = \min(2, 1) = 1$$
$$\{0, 1\}, \{2\} \to \{0, 1, 2\}; \quad \{7, 8, 9\} \text{ forms similarly}$$

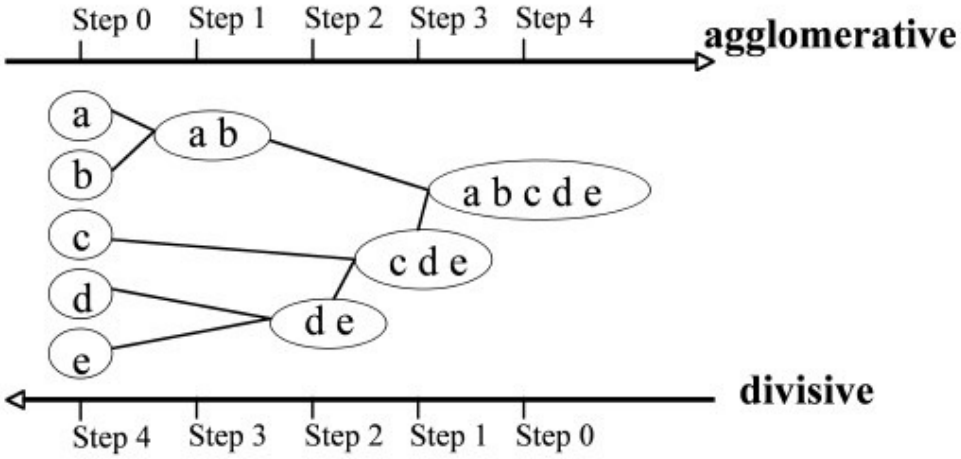
Work through it

The two triples finally merge at height 5, the closest cross-gap. Cutting at any height between 1 and 5 returns the two triples.

Interpretation

Early merges are retained. An incorrect similarity rule is not repaired by making the tree deeper.

Agglomeration and division traverse a hierarchy in opposite directions



Different linkage rules change the intermediate partition

Question and idea

Merge 0 and 1 first. Their group is distance 1 from point 2 under single linkage but distance 2 under complete linkage. Apply the same logic to 7,8,9.

$$d_{\text{single}}(\{0,1\},2) = \min(2,1) = 1$$
$$d_{\text{complete}}(\{0,1\},2) = \max(2,1) = 2$$

Calculate and explain

At cut height 1.5, single linkage gives two triples. Complete linkage has four groups under the same tie order:

$$\{0,1\}, \quad \{2\}, \quad \{7,8\}, \quad \{9\}$$

Meaning

A cut value is not comparable without its linkage definition. Equal raw coordinates do not imply identical hierarchy at every resolution.

Ward linkage minimizes the increase in within-group SSE

Idea

Ward asks how much squared error increases when two groups share a new mean. It favors merges that lose little compactness and should be used with a compatible Euclidean representation.

$$\Delta J(A, B) = \frac{|A||B|}{|A| + |B|} \|\mu_A - \mu_B\|^2$$

$$A = \{0, 1\},\ B = \{2\}: \quad \Delta J = \frac{2 \cdot 1}{3}(0.5 - 2)^2 = 1.5$$

Work through it

The pair $\{0, 1\}$ has SSE 0.5; the merged triple around mean 1 has SSE 2. Its increase is exactly 1.5.

Interpretation

Ward is a hierarchical compactness criterion. It does not reproduce every fixed-K K-means optimum, and a generic non-Euclidean distance is not interchangeable here.

Ward merge cost depends on sizes as well as mean separation

Class exercise

Compare two pairs of groups whose means are 0 and 2. One pair contains a single row in each group; the other contains 10 rows per group. The between-mean distance is the same.

$$\Delta J_{1,1} = \frac{1 \cdot 1}{2}(2)^2 = 2$$

$$\Delta J_{10,10} = \frac{10 \cdot 10}{20}(2)^2 = 20$$

Answer and explanation

For the larger groups, many more observations must move toward a shared mean, so the increase in within-group SSE is tenfold. Existing internal SSE cancels in the merge-increase formula.

Meaning

Ward height is an objective-change quantity, not merely a center distance. Implementation-specific plotted scaling still needs to be stated.

Read a dendrogram through merge heights and cuts

Read the figure

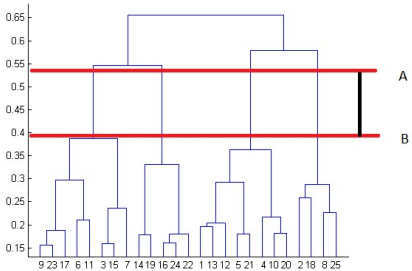
Leaves represent observations, branches record merges, and vertical heights encode the chosen linkage. A horizontal cut intersects branches to produce a partition at that resolution.

Follow the evidence

Count intersected branches at two cut heights. Locate large vertical gaps, then check whether the resulting groups make sense in original features and remain stable under small perturbations.

Interpretation

Horizontal spacing and branch rotation do not define sample distances. A large visual gap is a clue, not a universal K-selection theorem.



Two visibly different cuts can retain the same four groups

Question and idea

In the reference dendrogram, the indicated cuts near 0.53 and 0.395 lie between the lower four subtree merges and their next higher merges. Count crossed vertical branches rather than horizontal leaves.

$$\text{cut A} \approx .53: \quad 4 \text{ groups}$$
$$\text{cut B} \approx .395: \quad 4 \text{ groups}$$

Calculate and explain

The cuts have different heights but no merge event occurs between them. Hence the groups remain unchanged. A stable height interval is visible in the tree.

Meaning

This is a reading of the supplied reference figure, not a universal pair of threshold defaults. Inspect merge events and original-feature meaning before selecting a cut.

Hierarchy adds resolution but keeps its own limitations

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Idea

Single linkage can connect two clouds through a sparse chain. Complete and Ward may split elongated groups. All depend on representation, linkage, and early irreversible decisions.

$$\text{cloud A} \longleftrightarrow \text{sparse bridge} \longleftrightarrow \text{cloud B}$$

Work through it

A bridge of scores $3, 4, 5, 6$ inserted between our triples makes single-link distances small throughout. The obvious two-group gap disappears under that rule.

Interpretation

Pairwise storage often scales as N^2. For large datasets, sample carefully or choose methods whose computational structure matches the scale.

DBSCAN separates core points, border points, and noise

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Read the figure

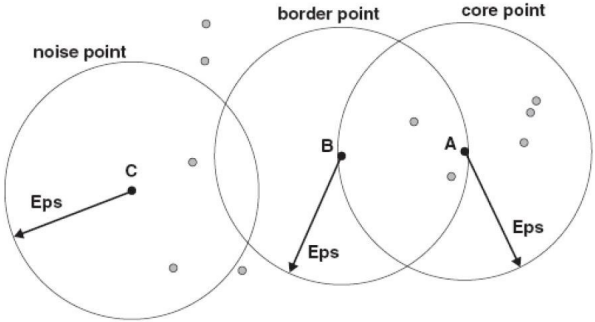
Specify radius ε and minimum neighbor count m. A core has enough rows in its neighborhood; a border is near a core but cannot expand; noise belongs to no accepted core-connected region.

Follow the evidence

State whether the row itself counts toward m; this lecture includes it. Core-to-core connections build a dense region, then reachable border rows join it.

Interpretation

Noise is an algorithmic role under chosen parameters, not proof of a faulty measurement. Border assignment can be ambiguous near two groups.



Translate minimum-count conventions before comparing implementations

Question and idea

Neighborhood counts include the query row. Excluding self preserves the same core rule if the threshold is lowered by one.

Role location	Including self	Excluding self
Triple center 1 or 8	3 neighbors, threshold 3	2 neighbors, threshold 2
Endpoint 0,2,7,9	2 neighbors	1 neighbor

Calculate and explain

Both conventions give two cores and four borders. Dropping self while retaining threshold 3 removes every core.

Meaning

A count parameter has meaning only with its counting convention and radius. Do not interpret different software outputs before reconciling those definitions.

Count neighborhoods on the same six-row example

Idea

Let $\varepsilon = 1.1$ and $m = 3$, including the row itself. Neighbor counts follow directly from our distance matrix and determine the point roles.

x	0	1	2	7	8	9
Count	2	3	2	2	3	2
Role	border	core	border	border	core	border

Work through it

Core 1 connects to borders $0, 2$; core 8 connects to borders $7, 9$. The two regions remain separate because their distance gap exceeds the radius.

Interpretation

Add an isolated score 20: it has one neighbor (itself), is near no core, and is therefore noise under these parameters.

Adding one row can turn an existing border into a core

Class exercise

Append score 3 to the six original scores and keep radius 1.1, minimum count 3 including self. Recompute every affected neighborhood instead of preserving old roles.

$$X = (0, 1, 2, 3, 7, 8, 9)$$

$$\text{counts} = (2, 3, 3, 2, 2, 3, 2), \quad \text{cores} = \{1, 2, 8\}$$

Answer and explanation

Score 2 used to have only 1,2 in its neighborhood; it now has 1,2,3 and can expand. Score 3 becomes a border reached from core 2, extending the lower group.

Meaning

Core and border are conditional density roles, not intrinsic categories of the objects. Data changes can alter both roles and connected regions.

Expand through core connections, not through every nearby point

Idea

Start from an unvisited core, join its neighborhood, and continue through newly found cores. Border rows are accepted but do not launch further expansion.

$$N_\varepsilon(x) = \{z : d(x,z) \le \varepsilon\} \qquad |N_\varepsilon(x)| \ge m \Rightarrow x \text{ is core}$$

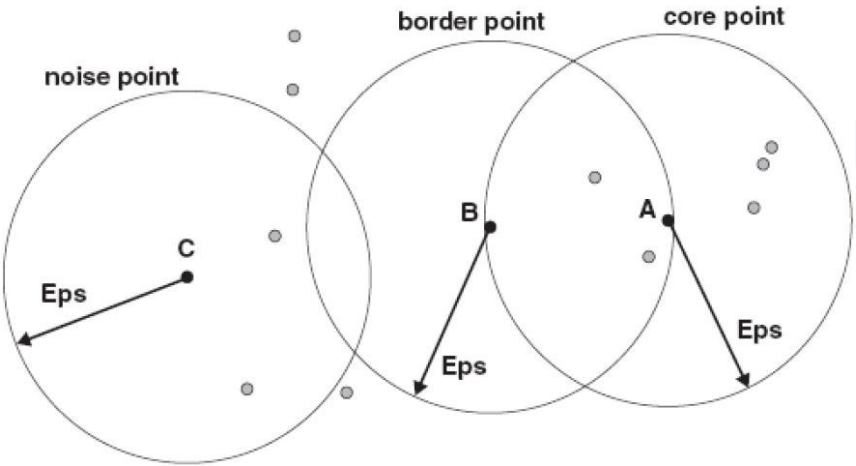
Work through it

In our example, point 1 starts the first region and reaches $0, 2$. Those borders do not extend it toward distant rows. Core 8 starts the second region independently.

Interpretation

Dense reachability is not ordinary transitive closeness through arbitrary sparse bridges. Distinguish the core graph from attached border rows.

Expand through core connections, not through every nearby point



Direct density reachability can be directional in symmetric geometry

Question and idea

In the triple 0,1,2, radius 1.1 and minimum count 3 make 1 a core and 0,2 borders. Point 0 is directly density-reachable from 1; reversing the direction requires 0 to be a core.

$$d(0,1) = d(1,0) = 1, \qquad \text{1 core}, \quad \text{0 not core}$$

Calculate and explain

The distance relation is symmetric, but the expansion permission is not. Both borders can still belong to the same density-connected group through common core 1.

Meaning

Distinguish near, directly reachable, and connected through a core path. Treating all three as ordinary undirected adjacency misstates DBSCAN's mechanism.

Radius and neighbor count control different thresholds

Two controls

Radius creates neighborhood links; minimum count asks how much local evidence is enough. Neither directly specifies the number of groups.

Parameters	Six scores, counting each row itself
$\varepsilon = 0.5, m = 3$	No core: every row is noise.
$\varepsilon = 1.1, m = 3$	Two groups, each with one core and two borders.
$\varepsilon = 5.1, m = 3$	The 2–7 gap becomes core-connected: one group.

A diagnostic

A sorted k-neighbor distance curve can suggest a radius transition; read its metric, count convention, and density distribution.

DBSCAN ε is an input-distance radius. SVR ϵ was an output-error tolerance: the shared symbol has different units and meaning.

A populated bridge can also connect density clusters

Question and idea

Add scores 3,4,5,6 to the two triples, creating every integer from 0 to 9. With radius 1.1 and minimum count 3 including self, interior bridge points become cores too.

$$X = \{0, 1, \ldots, 9\}, \quad |N_{1.1}(x)| = 3 \text{ for } x = 1, \ldots, 8$$

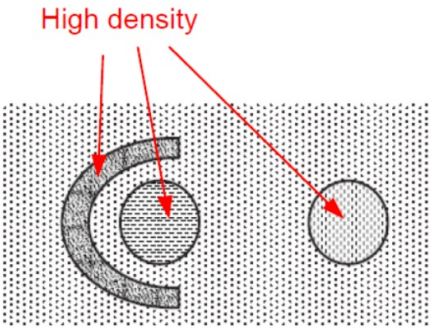
Calculate and explain

DBSCAN now forms one core-connected group with two endpoint borders, just as single-link connectedness follows the continuous chain. Raising the count to 4 would remove every core in this one-dimensional example.

Meaning

Density criteria do not guarantee that every bridge is rejected. Whether the bridge meets the chosen density definition must be checked, not assumed from appearance.

Density clusters are separated by low-density regions



(d) Density-based clusters. Clusters are regions of high density separated by regions of low density.

Relate a k-distance curve to the chosen count convention

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Class exercise

With minimum count 3 including self, a core needs at least two other neighbors. Compute distance to the second other neighbor for the original six rows and an isolated score 20.

$$d_{(2),\text{other}} = (2, 1, 2, 2, 1, 2, 12)$$
$$\text{sorted} = (1, 1, 2, 2, 2, 2, 12)$$

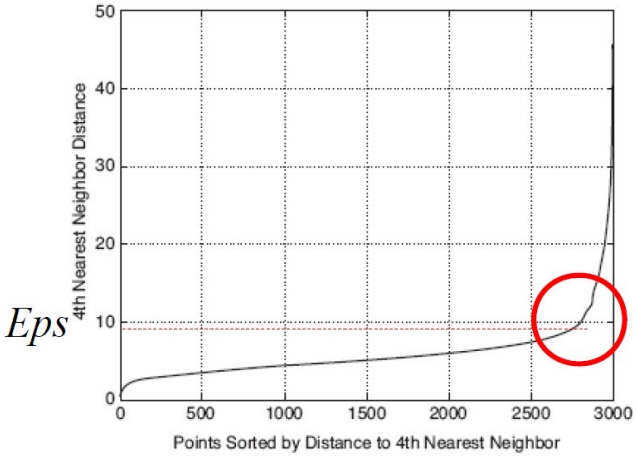
Answer and explanation

A radius around 2.1 makes the original six rows cores while leaving 20 without enough nearby rows. Radius 1.1 still creates the same two groups with only 1 and 8 as cores.

Meaning

The curve suggests a density scale, not a requirement that every grouped row be a core. Border rows can join at a narrower radius.

Relate a k-distance curve to the chosen count convention



A single radius struggles with unequal densities

Unequal density creates conflicting scales

Connecting a sparse group needs a wider radius; separating two nearby dense groups needs a narrower one.

$$A = \{0, .1, .2\}, \quad B = \{.6, .7, .8\}, \quad C = \{3, 4, 5\}, \quad m = 3$$

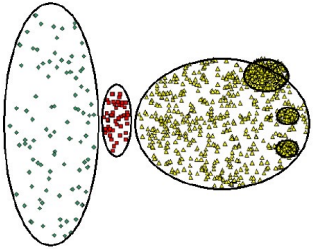
Radius	Result, counting each row itself
$\varepsilon = .15$	A and B form separate core-centered groups; C is entirely noise.
$\varepsilon = 1.1$	A and B merge through cores; sparse C forms its own group.

Explain the tradeoff

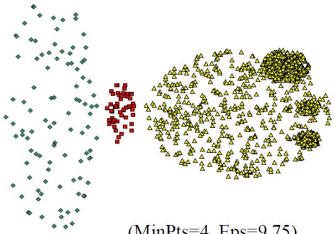
At radius .15, centers .1, .7 have three neighbors. At radius 1.1, score 4 becomes a core, but dense A and B are already connected.

One global radius cannot satisfy both desired resolutions here. Consider multiscale density or another representation.

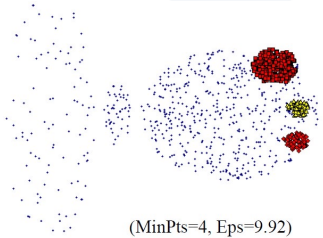
Unequal density makes a single neighborhood radius difficult to choose



Original Points



(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)

Rescale the radius when an input coordinate changes units

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Question and idea

Multiply every activity coordinate by 10. Pairwise distances multiply by 10, so preserving the same neighborhoods requires the radius to multiply by 10 while the count threshold stays fixed.

$$X' = (0, 10, 20, 70, 80, 90), \quad \varepsilon' = 10(1.1) = 11$$

Calculate and explain

Radius 11 recreates two cores and four borders with minimum count 3. Retaining radius 1.1 instead leaves each row alone and produces no core.

Meaning

A density threshold is tied to input units. Anisotropic changes across several features require reconsidering the metric, not only multiplying one global radius.

A similarity graph turns grouping into a connectivity question

Idea

Represent each row as a node and connect sufficiently similar rows with weighted edges. A graph partition seeks internally strong connections with relatively weak connections between groups.

$$W_{ij} = \exp(-\|x_i - x_j\|^2/(2\sigma^2)) \quad \text{or a symmetric neighbor graph}$$

$$D_{ii} = \sum_j W_{ij}, \qquad L = D - W$$

Work through it

For our six scores with only radius-1.1 edges, the graph contains paths 0–1–2 and 7–8–9. Their disconnected structure supports two components.

Interpretation

Graph construction is a modeling choice: width, neighbors, symmetry, and scaling determine which structure a later cut can discover.

Compare a weighted graph with a thresholded neighbor graph

Question and idea

For the six scores, use RBF edge weight $e^{-0.5d^2}$ and omit self-loops. Nearby within-triple edges are strong; the cross-gap edge is tiny but still nonzero.

$$d = 1: \ W = .607; \quad d = 2: \ W = .135; \quad d = 5: \ W = e^{-12.5} \approx 3.73 \times 10^{-6}$$

Calculate and explain

A full RBF graph remains connected. A radius 1.1 threshold deletes all cross-gap edges and gives exactly two disconnected paths. The two graph constructions imply different spectra.

Meaning

Nonnegative similarity is enough for this graph construction; zeroing diagonals or edges does not need to preserve a kernel Gram interpretation.

Spectral clustering embeds connectivity before partitioning

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Idea

Use selected eigenvectors of a graph Laplacian to represent each node, then partition these new row vectors. The spectral coordinates describe graph connection patterns rather than raw Euclidean position.

$$L_{\mathrm{sym}} = I - D^{-1/2} W D^{-1/2}$$

$$L_{\mathrm{sym}} u_k = \lambda_k u_k \quad \longrightarrow \quad U = [u_1, \ldots, u_K] \quad \longrightarrow \quad \text{cluster rows of } U$$

Work through it

Disconnected components give zero eigenvalues and component-indicator structure. Weak bridges perturb this into approximately separated coordinates, potentially revealing nonconvex groups.

Interpretation

Choose a graph and embedding convention explicitly. Eigenvalue gaps are clues, while graph quality, scaling, cost, and stability still need evaluation.

Recover the six-row groups from two Laplacian zero modes

Class exercise

The radius-1.1 graph has unit edges $(0,1),(1,2),(7,8),(8,9)$ and no self-loops. Each triple is a path. Its Laplacian block uses endpoint degree 1 and middle degree 2.

$$B = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \qquad L = \operatorname{diag}(B, B)$$

$$u_1 = (1,1,1,0,0,0)^T, \quad u_2 = (0,0,0,1,1,1)^T, \quad Lu_1 = Lu_2 = 0$$

Answer and explanation

The spectrum is $(0,0,1,1,3,3)$. For L_{sym}, zero vectors are $D^{1/2}u_k$; middle coordinates become $\sqrt{2}$. Row normalization then maps each triple to one common component direction.

Meaning

Two zero modes identify two disconnected components. A weak cross-edge leaves one exact zero mode and one small positive eigenvalue.

Select the structural question before selecting the package

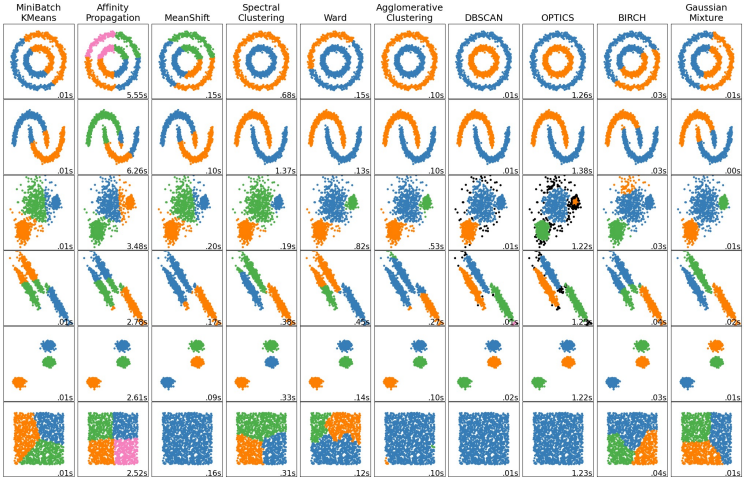
Match definition to evidence

Choose a structural question first, then a method and its assumptions.

Method	Starting evidence	Qualification
K-means	Compact center-based groups	Scale, outliers, local minima
GMM	Elliptical overlap; soft membership	Covariance and density interpretation
Hierarchy	Useful nested resolutions	Linkage, bridges, storage
DBSCAN	Dense irregular groups plus noise	Radius and unequal densities
Spectral	Strong within-graph connections	Graph construction and computation

Our six compact scores suit several methods. Add overlap, a sparse bridge, or an isolated row to expose their differences. Compare original features, stability, and whether the groups warrant different actions.

Select the structural question before selecting the package



Explain one complete grouping analysis and its limits

Idea

For activity-based outreach, use one row per customer and a stated time window. Start with the six scores, choose an interpretable metric, and compare simple structural models.

1. K-means with seeds $0, 7$: assignments, means $1, 8$, SSE $10 \to 4$.
2. At score 1: silhouette $6/7 \approx 0.857$.
3. GMM with means $1, 8$ and variances 9: soft responsibilities, then means $1.440, 7.560$.
4. DBSCAN with radius 1.1, count 3: two cores and four borders; isolated 20 is noise.

Work through it

Explain the groups in original behavior terms, repeat sensible perturbations, and determine whether different actions are warranted.

Interpretation

Next lecture changes representation through dimensionality reduction. Any newly separated plot must still be evaluated against the original question.

A fixed K may spend a center on an isolated observation

Class exercise

Add activity 20 to the six original scores and keep $K = 2$. Compare two feasible partitions before declaring that the old low/high interpretation must survive.

$$P_A = \{0, 1, 2 \mid 7, 8, 9, 20\}: \quad \mu = (1, 11), \quad J = 2 + 110 = 112$$
$$P_B = \{0, 1, 2, 7, 8, 9 \mid 20\}: \quad \mu = (4.5, 20), \quad J = 77.5$$

Answer and explanation

The second candidate has lower SSE even though it merges the old low/high groups. With three centers, the original triples plus singleton 20 have SSE 4; DBSCAN can instead retain two dense groups and mark 20 as noise.

Meaning

Outputs reflect different grouping questions. A fixed number of complete groups is not interchangeable with two dense groups plus an unassigned outlier.